

Vinaykumar Venkateshkumar

linkedin.com/in/vinaykumar-venkateshkumar
github.com/Itsvid · vinaykumar.is-a.dev
Cranfield, United Kingdom

kumarvsvinay@gmail.com
+44 7742 914241

Work authorisation: sponsorship required
Available: September 2026

Design engineer with an aerospace propulsion background. Produces release-ready mechanical design work — dimensioned detail and general arrangement drawings with geometric tolerancing to ISO 1101, limits and fits to ISO 286, tolerance stack-up, sheet-metal flat patterns and design-for-manufacture checks — from parametric CAD models built in CATIA V5, CadQuery and pyOCC, and sizes them against load cases with finite-element analysis. Verifies geometry rather than assuming it, and treats a simulation result as a claim to be tested rather than an answer. MSc Thermal Power and Propulsion at Cranfield. Seeking an entry-level design engineering role in aerospace or propulsion hardware.

TECHNICAL SKILLS

Mechanical Design	Geometric tolerancing (ISO 1101), limits and fits (ISO 286), tolerance stack-up (worst-case and RSS), design for manufacture, sheet-metal forming (bend allowance, flat patterns), detail and GA drawings, BOM generation
CAD	CATIA V5 (proficient), CadQuery, pyOCC / Open CASCADE, ANSYS DesignModeler, STEP and DXF interchange, parametric geometry (iCST)
Structural Analysis	Finite-element analysis (CalculiX, gmsh, FreeCAD FEM), static load cases, mesh convergence studies, identifying stress singularities, hand calculation as an independent check
CFD & Simulation	ANSYS Fluent, ANSYS ICEM CFD / Meshing, OpenFOAM, $k-\epsilon$ / $k-\omega$ turbulence modelling, grid-independence studies
Programming	Python (NumPy, SciPy, Matplotlib, pytest), MATLAB, Git, automated test suites
Engineering Domains	Gas turbine performance, propulsion, turbomachinery aerodynamics, thermodynamics, aircraft structures, non-destructive testing

DESIGN PROJECTS

- **Parametric Actuator and Gearbox Generators with Drawing Pack — (Mechanical design, GD&T, CadQuery):** Wrote two parametric generators that size a design before drawing it: a hydraulic flight-control actuator family across four aircraft classes, and an engine accessory gearbox across five power ratings (5–50 kW), with involute gear geometry, Lewis bending-stress module sizing and SKF 60xx bearing selection. Produced a four-sheet A4 drawing pack — three detail drawings and a ballooned assembly GA — carrying ISO 286 fits (Ø35 H8 bore, Ø21 f7 rod), ISO 1101 geometric tolerances, a chosen datum scheme, surface finishes, a parts list and a five-contributor tolerance stack (390 mm nominal, worst case ± 0.90 , RSS ± 0.44). Verification found six real defects, including a housing that did not enclose its own gear, and meshing teeth that physically intersected by 424 mm^3 in every exported assembly because the two gears were never phased — fixed by offsetting one by half a tooth pitch, which is parity-dependent and wrong to apply to an odd gear. Checked the test suite by reintroducing a known sign error: it is caught only by testing that the tooth narrows toward its tip, while pitch-circle thickness and solid-validity checks both still pass. 132 automated tests. *Tech: Python, CadQuery, OpenCASCADE, ISO 286, ISO 1101, STEP*
- **Formed Sheet-Metal Equipment Bracket — (Sheet-metal design, DFM):** Parametric formed bracket with its flat pattern derived from a K-factor neutral axis, not assumed — 101.52 mm blank against 105.00 mm of summed outside legs. Validated by conservation of volume: the formed solid and the flat blank agree to 0.25%, against 5.12% for a naively summed blank — a bound tightened from 2% after the looser one was found to be hiding two missing holes. Five design-for-manufacture rules (minimum bend radius, flange length, hole-to-bend distance, fastener edge distance, minimum hole diameter) with a material trade that has a real answer: 2024-T3 is twice as strong as 5052-H32 and cannot make the fold, needing 4T minimum bend radius against 1T. Detail drawing with bend table; STEP and DXF export. *Tech: Python, CadQuery, sheet-metal forming, DFM*
- **Structural Sizing and Redesign of That Bracket — (FEA, mesh convergence, design trade study):** Sized the bracket against a 9g equipment crash load in CalculiX, driven headlessly through FreeCAD. Ran a five-density mesh convergence study rather than a single solve, which is what made the result usable: deflection converged to 0.06% and bulk stress to 0.28%, while peak stress climbed 12% and never settled — a singularity at the edge of a fixed constraint, so the 545 MPa headline is a boundary condition rather than a stress. The converged 196.7 MPa against a 193 MPa yield condemns the part marginally, not by the $2.8\times$ a single mesh reports. Screened four fixes analytically before modelling any: stiffening flanges up the upright make that leg $7.4\times$ stronger and change the margin by nothing, because both legs carry the same moment through the same section and the governing stress moves to the untouched base. Selected 2.0 mm gauge, solved it to confirm (126.1 MPa, +53% margin, deflection halved), and re-issued the drawing — the bend deduction changes with gauge, so a shop cutting the old blank would make the wrong part. Cross-checked against closed-form beam theory sharing no code, which reads 16% conservative at both gauges to within 0.2 of a point. 66 automated tests. *Tech: CalculiX, gmsh, FreeCAD FEM, Python, CadQuery*

- **Installation Aerodynamics of Aero-Engine Nacelles — Cranfield Individual Research Project (Parametric CAD, CFD):** Fitted a Class-Shape Transformation curve directly to NASA TM 110300's published NACA 1-85-100 cowl ordinates, reproducing them to 0.019% RMS of maximum radius. Built and converged a compressible RANS OpenFOAM case (rhoSimpleFoam, $k-\omega$ SST) at Mach 0.79, meshed with snappyHexMesh at 108,904 cells, and compared surface pressure against the same report's wind-tunnel data — RMS C_p error 0.65 over 17 matched stations, reported including where the model misses the leading-edge suction peak. *Tech: pyOCC, Open CASCADE, iCST, OpenFOAM, Python (Ongoing, expected Sep 2026)*
- **Twin-Spool Turbofan Cycle Model — (Gas turbine performance):** Station-by-station separate-exhaust cycle solver with every turbine pressure ratio matched to the power its own spool demands rather than assumed. 57.67 kN net thrust, 46.7% thermal efficiency at the reference design point, validated against the closed-form ideal-Brayton limit to within 0.01%. 82 tests. *Tech: Python, Matplotlib, pytest, MATLAB/Simulink cross-check*
- **Investigation of Controlled Jets for Enhanced Mixing Rates — BEng Final Year Project, team of 3 (CFD, experimental validation):** Investigated passive flow-control tab geometries (Delta Tandem Tab, M Delta Tandem Tab) for jet mixing using ANSYS Fluent across Mach 0.6–1.0, with grid-independence studies to 3.4M+ nodes, validated against experimental shadowgraph imaging. Achieved 64–76% core-length reduction against an uncontrolled baseline at Mach 0.8 with no significant thrust penalty. *Tech: ANSYS Fluent, ANSYS DesignModeler, ANSYS ICEM CFD (Mar 2025)*

Further work at vinaykumar.is-a.dev: airfoil panel method written from scratch and cross-checked against XFOil (lift agreeing to 0.047 RMS), flight-performance calculator validated against three aircraft, OpenFOAM NACA 0012 study (GCI 6.4%), parametric wing and compressor blade-row generators.

EDUCATION

Cranfield University

MSc Thermal Power and Propulsion

Modules: Gas Turbine Performance, Combustion, Turbomachinery Aerodynamics, Propulsion System Design

Research project (ongoing): Installation aerodynamics of aero-engine nacelles

Cranfield, UK

Oct 2025 – Sep 2026

KCG College of Technology, Anna University

BEng Aeronautical Engineering — CGPA 7.37/10

Courses: Air Breathing Propulsion, Rocket Propulsion, Aero Engineering Thermodynamics, Aerodynamics I & II, Computational Fluid Dynamics, Aircraft Structures I & II, Finite Element Methods, Wind Tunnel Techniques, Non-Destructive Testing, Aircraft Design

Chennai, India

2021 – 2025

EXPERIENCE

AI Engineering Services Limited (AIESL) — Boeing 737 MRO Base

Engineering Intern, Aircraft Maintenance

Thiruvananthapuram, India

Mar 2025 – Apr 2025

- Rotational internship across Component Overhaul, Material and Production Planning, and Stores at a 737 base-maintenance facility servicing Air India Express.
- Documented wheel and brake overhaul procedures to specification (torque 158 lb-ft, tyre pressure 205 ± 5 psi) and Eddy Current Testing for non-destructive flaw detection.
- Used AMOS and RAMCO systems to track parts issuance, stock levels and task-card compliance.

CERTIFICATIONS AND ACHIEVEMENTS

- CATIA V5 (Proficient) — CADD Centre Training Services, 80 hrs — 2024
- IELTS Academic — Overall Band 6.5 (L 7.0, R 6.5, W 6.5, S 5.5); UKVI B2 — Jan 2025
- MATLAB Essential Training — LinkedIn Learning — 2024
- Machine Learning with Python; Deep Learning with TensorFlow — IBM / Cognitive Class — 2024